CLASSIFICATION

INTRODUCTION

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INTRODUCTION

The following concepts will be introduced:

- ✓ **EXPLANATORY/CLASS ATTRIBUTE**
- ✓ **DESCRIPTIVE/PREDICTIVE MODELING**
- ✓ **TRAINING SET/TEST SET**
- ✓ **CLASSIFICATION MODEL (LEARNER – INDUCER)**
- ✓ **CONFUSION MATRIX**
- ✓ **ACCURACY/ERROR**

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Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	n	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	y	7	1.78	OH	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	n	6	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	510	355-9993
147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173

The **CUSTOMERS CALLS DATABASE**, provided by the **CEO OF THE TELECOM COMPANY**, contains customers' call data consisting of many attributes including **CHURN**, a binary attribute representing whether a given **CUSTOMER CHURNED FOR A COMPETITOR OR NOT**.

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The **CUSTOMERS CALLS DATABASE** consists of more than **3K RECORDS** and **21 FIELDS**:

- **STATE** STATE WHERE THE CUSTOMER LIVES {KS, OH, NJ, …}
- **INTL CALLS** NUMBER OF INTERNATIONAL CALLS
- **DAY MINS** MINUTES OF CALLS DURING THE DAY
- **PHONE** PHONE NUMBER
- **…**
- **CHURN** HAS THE CLIENT CHURNED? {Y, N}

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The **Customers Calls Database** consists of more than **3K records** and **21 fields**:

- **State** **(Categorical – nominal)** State where the customer lives {KS, OH, NJ, …}
- **Intl Calls** **(Numeric – interval)** number of international calls
- **Day Mins** **(Numeric – interval)** minutes of calls during the day
- **Phone** **(Categorical – nominal)** phone number
- **…**
- **Churn** **(Categorical – binary)** has the client churned? {Y, N}

The **CEO of the Telecom Company** would like to predict the **attribute Churn**.

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You make the decision to develop a **CLASSIFICATION MODEL**

- **INTL CALLS, DAY MINS, ...** ARE REFERRED TO AS **EXPLANATORY OR INPUT ATTRIBUTES**
- **CHURN** IS REFERRED TO AS THE **CLASS OR OUTPUT ATTRIBUTE**

You make the decision to develop a **CLASSIFICATION MODEL**

Intl Calls →
Day Mins →
... →
... →
Classification Model → Churn

- **Intl Calls, Day Mins, ...** are referred to as **Explanatory or Input Attributes**
- **Churn** is referred to as the **Class or Output Attribute**

Explanatory Attributes / Input Attributes → **Classification Model** → Class Attribute / Output Attribute

A **CLASSIFICATION MODEL** solves the **CLASSIFICATION TASK** or **CLASSIFICATION PROBLEM**. It is useful for the following purposes:

- ✓ **DESCRIPTIVE MODELING**: can serve as explanatory tool to distinguish between objects of different classes

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- ✓ **PREDICTIVE MODELING**: predicts the class of an unknown record. It can be treated as a BLACK BOX that automatically assigns a class label when presented with the attribute set of an unknown record.

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
74	0	187.7	?	5	2.46	RI	415	344-9403

EXPLANATORY ATTRIBUTES

INPUT ATTRIBUTES

CLASS ATTRIBUTE

OUTPUT ATTRIBUTE

Who gives us the CLASSIFICATION MODEL?

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A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.

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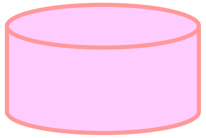

TRAINING SET

The **TRAINING SET (TRAINING DATA SET)** consists of records whose **CLASS LABELS** (class values) **ARE KNOWN**.

The Training set must be provided.

It is **USED TO BUILD** (learn and validate) a **CLASSIFICATION MODEL**, which is subsequently **APPLIED** to the **TEST SET**.

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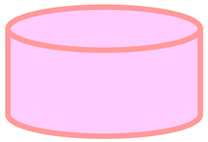
A **Classification Technique (Classifier)** is a systematic approach to building **Classification Models** from a **Data Set**.

Training Set



Test Set

The **Test Set (Test Data Set)** consists of records whose **class labels** (class values) **are unknown** (assumed to be such).

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The **CLASSIFICATION MODEL** is **LEARNT USING THE** records of the **TRAINING SET**.

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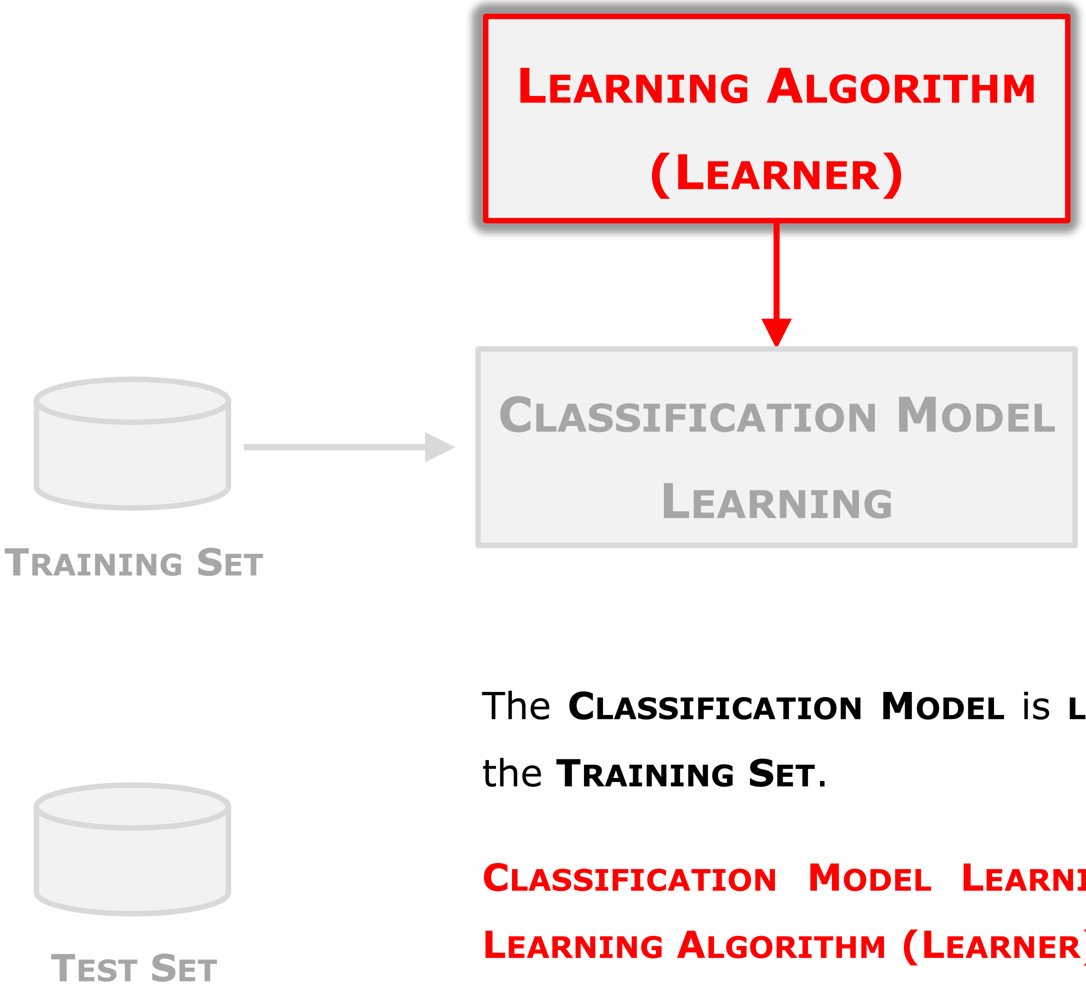

The **CLASSIFICATION MODEL** is **LEARNT USING THE** records of the **TRAINING SET**.

CLASSIFICATION MODEL LEARNING takes place using the **LEARNING ALGORITHM (LEARNER)**.

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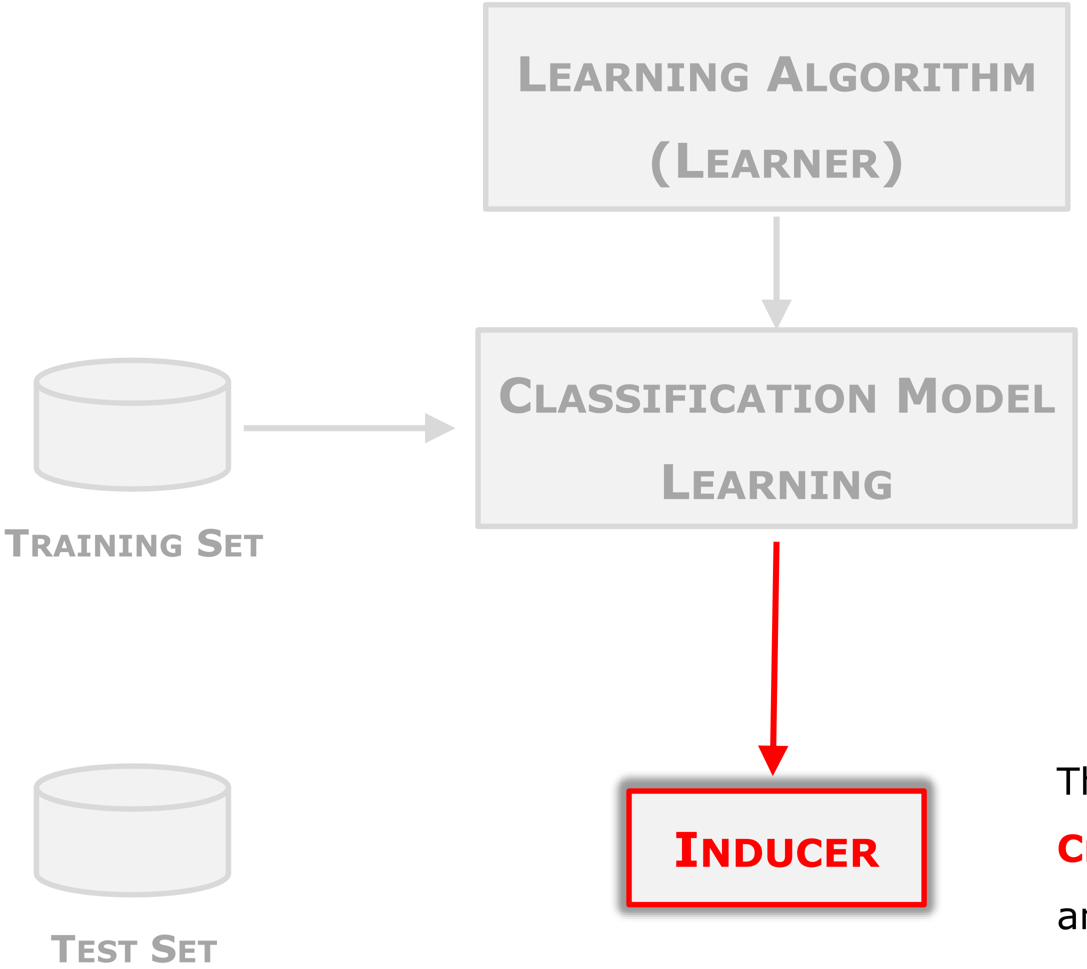
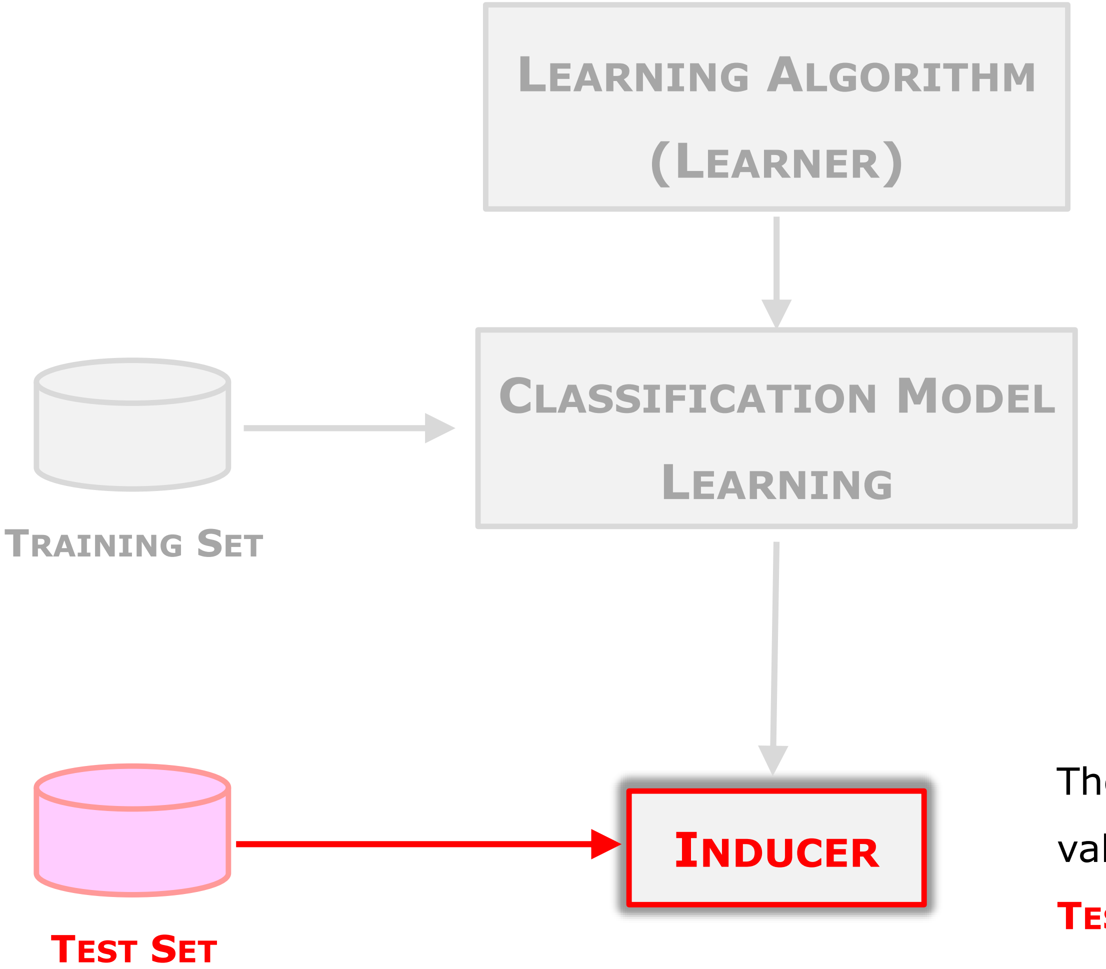


The **INDUCER** is the output of the **CLASSIFICATION MODEL LEARNING**, it is an **INSTANCE OF THE CLASSIFICATION MODEL**.

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.



The **INDUCER** is queried to **PREDICT** the value of **THE CLASS ATTRIBUTE** for the **TEST SET RECORDS**.

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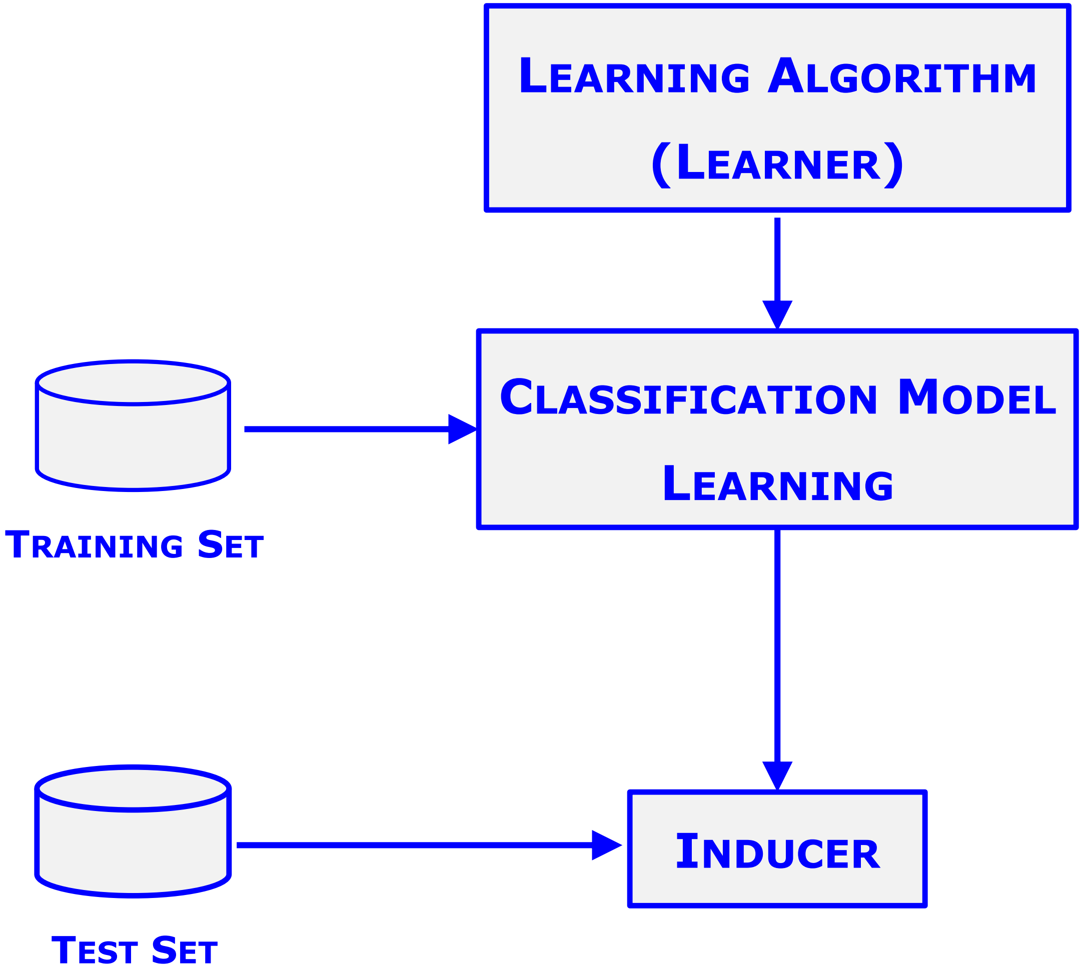
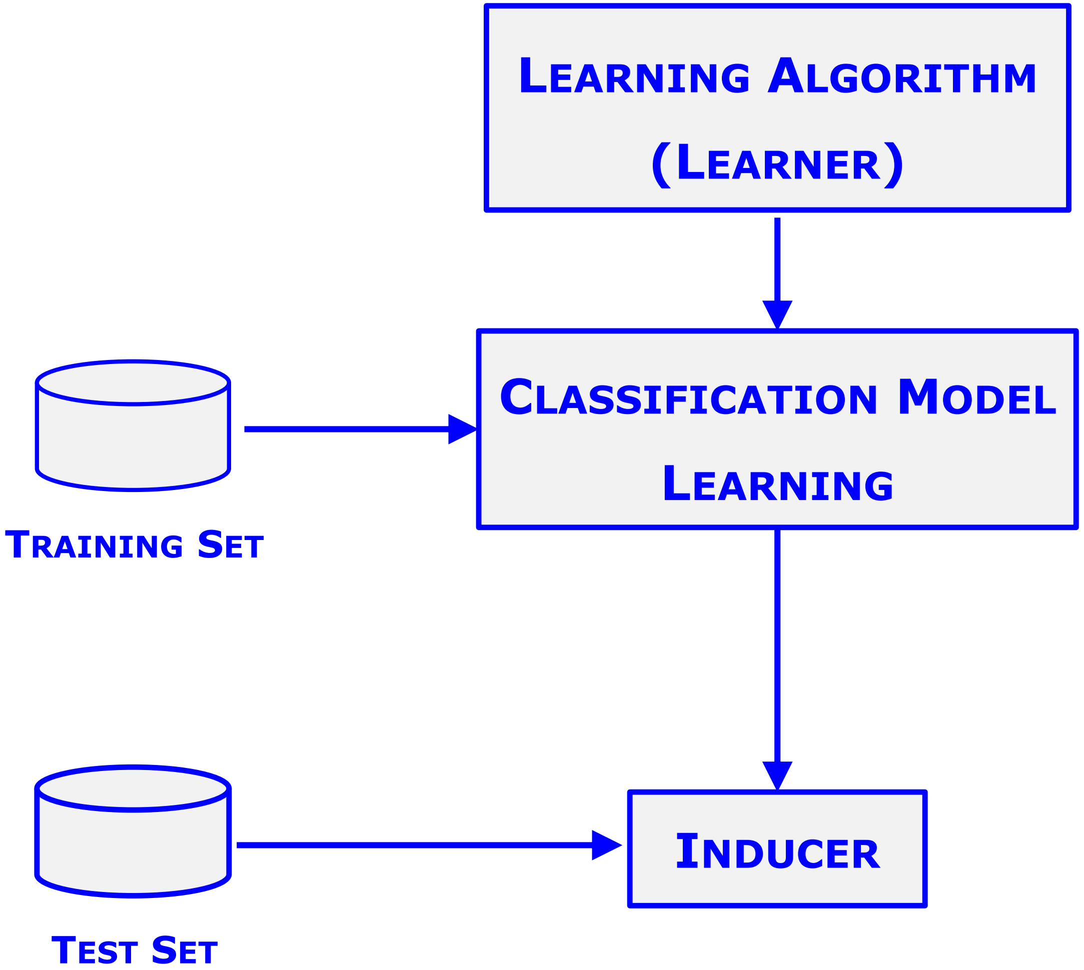


This schema summarizes the **general approach to solving a Classification Problem**.

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.



This schema summarizes the **GENERAL APPROACH TO SOLVING A CLASSIFICATION PROBLEM.**

How to measure the PERFORMANCE OF THE CLASSIFICATION MODEL?

PERFORMANCE EVALUATION of a **CLASSIFICATION MODEL** is based on the **COUNTS OF TEST RECORDS CORRECTLY AND INCORRECTLY PREDICTED BY THE INDUCER**.

These counts are tabulated in a table known as **CONFUSION MATRIX**.

		INDUCER PREDICTION (IP)	
		-1	+1
ACTUAL CLASS (AC)	-1	TN	FP
	+1	FN	TP

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✓ **TN, true negative**, # of records where **AC=-1** which are correctly predicted **IP=-1**

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		INDUCER PREDICTION (IP)	
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- ✓ **TN**, **true negative**, # of records where **AC=-1** which are correctly predicted **IP=-1**
- ✓ **FN, false negative,** # of records where **AC=+1** which are wrongly predicted **IP=-1**

Performance evaluation of a **Classification Model** is based on the **counts of test records correctly and incorrectly predicted by the Inducer**.

These counts are tabulated in a table known as **Confusion Matrix**.

		Inducer Prediction (IP)	
		-1	**+1**
Actual Class (AC)	-1	TN	FP
	+1	FN	**TP**

- ✓ **TN**, **true negative**, # of records where **AC=-1** which are correctly predicted **IP=-1**
- ✓ **FN, false negative,** # of records where **AC=+1** which are wrongly predicted **IP=-1**
- ✓ **TP, true positive**, # of records where **AC=+1** which are correctly predicted **IP=+1**

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PERFORMANCE EVALUATION of a **CLASSIFICATION MODEL** is based on the **COUNTS OF TEST RECORDS CORRECTLY AND INCORRECTLY PREDICTED BY THE INDUCER.**

These counts are tabulated in a table known as **CONFUSION MATRIX.**

		INDUCER PREDICTION (IP)	
		-1	**+1**
ACTUAL CLASS (AC)	**-1**	TN	**FP**
	+1	FN	TP

- ✓ **TN**, **true negative**, # of records where **AC=-1** which are correctly predicted **IP=-1**
- ✓ **FN, false negative,** # of records where **AC=+1** which are wrongly predicted **IP=-1**
- ✓ **TP, true positive,** # of records where **AC=+1** which are correctly predicted **IP=+1**
- ✓ **FP, false positive,** # of records where **AC=-1** which are wrongly predicted **IP=+1**

The Confusion Matrix contains the information needed to evaluate the performance of the Classification Model.

Summarizing this information with a single number would make it more convenient to **COMPARE THE PERFORMANCE OF DIFFERENT CLASSIFICATION MODELS**.

		INDUCER PREDICTION (IP)	
		-1	+1
ACTUAL CLASS (AC)	-1	**TN**	FP
	+1	FN	**TP**

This can be done using a **PERFORMANCE METRIC** such as **ACCURACY**, which is defined as follows:

NUMBER OF CORRECT PREDICTIONS

$$Accuracy = \frac{\boxed{TN + TP}}{TN + FN + FP + TP}$$

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$$Accuracy = \frac{TN + TP}{\boxed{TN + FN + FP + TP}}$$

TOTAL NUMBER OF PREDICTIONS

Introduction

The Confusion Matrix contains the information needed to evaluate the performance of the Classification Model.

Summarizing this information with a single number would make it more convenient to **compare the performance of different Classification Models**.

		Inducer Prediction (IP)	
		-1	+1
Actual Class (AC)	-1	TN	**FP**
	+1	**FN**	TP

A complementary view is provided by using as **Performance Metric** the **Error**, which is defined as follows:

Number of wrong predictions

$$Error = \frac{\boxed{FN + FP}}{TN + FN + FP + TP}$$

The Confusion Matrix contains the information needed to evaluate the performance of the Classification Model.

Summarizing this information with a single number would make it more convenient to **COMPARE THE PERFORMANCE OF DIFFERENT CLASSIFICATION MODELS.**

		INDUCER PREDICTION (IP)	
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ACTUAL CLASS (AC)	-1	**TN**	**FP**
	+1	**FN**	**TP**

A complementary view is provided by using as **PERFORMANCE METRIC** the **ERROR**, which is defined as follows:

NUMBER OF WRONG PREDICTIONS

$$Error = \frac{FN + FP}{\boxed{TN + FN + FP + TP}}$$

TOTAL NUMBER OF PREDICTIONS

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The Confusion Matrix contains the information needed to evaluate the performance of the Classification Model.

Summarizing this information with a single number would make it more convenient to **COMPARE THE PERFORMANCE OF DIFFERENT CLASSIFICATION MODELS.**

$$Accuracy = \frac{TN + TP}{TN + FN + FP + TP} \qquad Error = \frac{FN + FP}{TN + FN + FP + TP}$$

$$Error = 1 - Accuracy$$